Chapter 0 Exercises

Directions: In exercises 1 - 12, find the distance between the points. Write the answer in exact form and then find the decimal approximation, rounded to one decimal place if needed.

1. $(2,0)$ and $(5,4)$

2. $(-4,-3)$ and $(2,5)$

3. $(-4,-3)$ and $(8,2)$

4. $(-7,-3)$ and $(8,5)$

5. $(-1,4)$ and $(2,0)$

6. $(-1,3)$ and $(5,-5)$

7. $(1,-4)$ and $(6,8)$

8. $(-8,-2)$ and $(7,6)$

9. $(-3,-5)$ and $(0,1)$

10. $(-1,-2)$ and $(-3,4)$

11. $(3,-1)$ and $(1,7)$

12. $(-4,-5)$ and $(7,4)$

Directions: In exercises 13 - 16, find the midpoint of the line segments whose endpoints are given.

13. $(0,-5)$ and $(4,-3)$

14. $(-2,-6)$ and $(6,-2)$

15. $(3,-1)$ and $(4,-2)$

16. $(-3,-3)$ and $(6,-1)$

Directions: In exercises 17 - 20, write the standard form of the equation of the circle with the given radius and center

17. Radius: 1, center: $(3,5)$

18. Radius: 10, center: $(-2,6)$

19. Radius: 2.5, center: $(1.5,-3.5)$

20. Radius: 1.5, center: $(-5.5,-6.5)$

Directions: In exercises 21 - 32, ⓐ find the center and radius, then ⓑ graph each circle.

21. $(x+5)^2+(y+3)^2=1$

22. $(x-2)^2+(y-3)^2=9$

23. $(x-4)^2+(y+2)^2=16$

24. $(x+2)^2+(y-5)^2=4$

25. $x^2+(y+2)^2=25$

26. $(x-1)^2+y^2=36$

27. $(x-1.5)^2+(y+2.5)^2=0.25$

28. $(x-1)^2+(y-3)^2=\frac{9}{4}$

29. $x^2+y^2=64$

30. $x^2+y^2=49$

31. $2x^2+2y^2=8$

32. $6x^2+6y^2=216$

Directions: In exercises 33 - 40, ⓐ identify the center and radius and ⓑ graph.

33. $x^2+y^2+2x+6y+9=0$

34. $x^2+y^2-6x-8y=0$

35. $x^2+y^2-4x+10y-7=0$

36. $x^2+y^2+12x-14y+21=0$

37. $x^2+y^2+6y+5=0$

38. $x^2+y^2-10y=0$

39. $x^2+y^2+4x=0$

40. $x^2+y^2-14x+13=0$